

LECTURE-7

Monday Wednesday
9/9/26.

Floating Point:

Single Precision → 32 Bits

Double → 64 Bits

Binary → Scientific Notation → Single / double mapping for floating point no.

Q: $(38.75)_{10} = 3.875 \times 10^1$

2	38
19	-0
9	-1
4	-1
2	-0
1	-0

$$\begin{array}{r}
 0.75 \\
 \times 2 \\
 \hline
 1.50 - 1 \\
 \hline
 0.50 \\
 \times 2 \\
 \hline
 1.00 - 1
 \end{array}$$

$(100110.11)_2$

stop after 3-4 times

100110.11 scientific not. → 1.0011011×2^5

5 4 3 2 1
5+127

Bias for single precision
 $5+127 = (132)_{10}$
 $= (10000100)_2$

S	Exp	fraction
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+ve: 0
-ve: 1

Single: 1	Single: 8 Bits	Single: 23 bits	= 32 bits
Double: 1	Double: 11 Bits	Double: 52 Bits.	= 64 bits

because +ve

0	10000100	001101100000000000000000
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extra zeros: on left of 'exp' & right of 'fraction'

